

From Redlining to Today: HOLC Grades and Housing Inequality in the Bay Area

Emily Ma Larissa Arreola Moreno Vittal Vasudevan
Daniel Porter

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1 Introduction

Redlining refers to the practice of grading neighborhoods by perceived lending risk in the mid-twentieth century, most notably through maps produced by the Home Owners' Loan Corporation (HOLC) in the 1930s. These maps classified neighborhoods from “A” (Best) to “D” (Hazardous), with areas graded C and D disproportionately associated with communities of color and immigrant populations. Though redlining policies were outlawed decades ago, research shows that the contemporary housing market is still in the shadow of the redlining's spatial and economic consequences.

This project investigates whether patterns associated with historical redlining continue to be reflected in present-day housing and demographic outcomes. Using national HOLC data and county-level American Community Survey (ACS) estimates, the analysis situates redlining in historical context and evaluates its persistence in the Bay Area, focusing on Alameda (Oakland), Santa Clara (San Jose), and San Francisco counties.

1.1 Goal of Analysis

The main goal of this analysis is to assess whether historically redlined areas exhibit persistent disparities in housing and demographic composition. The project addresses three main questions:

1. How do housing value and demographics differ across HOLC grades at the national level?
2. Do similar patterns appear within Bay Area counties?
3. Is there evidence that disparities between redlined and non-redlined areas have narrowed over time?

1.2 Data and Methodological Overview

- **Historical Redlining (HOLC):** Digitized Home Owners' Loan Corporation (HOLC) neighborhood grades were obtained from the [Mapping Inequality Project](#), which provides spatial data on historical redlining practices in U.S. cities.
- **National Redlining Time Series:** National-level redlining-related housing indicators were accessed through the Federal Reserve Bank of St. Louis' [FRED database](#), which aggregates historical economic and housing time series.
- **Contemporary Housing and Demographics:** Tract-level housing and demographic data were drawn from the U.S. Census Bureau's [American Community Survey \(ACS\) 5-Year Estimates](#), covering the period 2009–2022. Included data table: Race(B03002), Income(B19013), Median Home Value(B25077), Homeownership(B25003)

Methodologically, the project combines spatial analysis, visualization and regression-based comparison. National time-series analyses provide historical context, while Bay area case studies link HOLC grades to current day outcomes. Difference-in-difference comparison and regression are used to examine whether the inequality persisted over time. the result are interpreted as associational evidence rather than casual effects.

2 National Context - Redlining as a structural Pattern

We begin by situating redlining within a national historical context. Figure 1 shows national trends in housing values across HOLC grades over time. Across the entire period, neighborhoods graded C and D consistently exhibit lower median home values compared to Grade A neighborhoods. Importantly, there is little visual evidence that this gap narrows over time, suggesting persistent disparities.

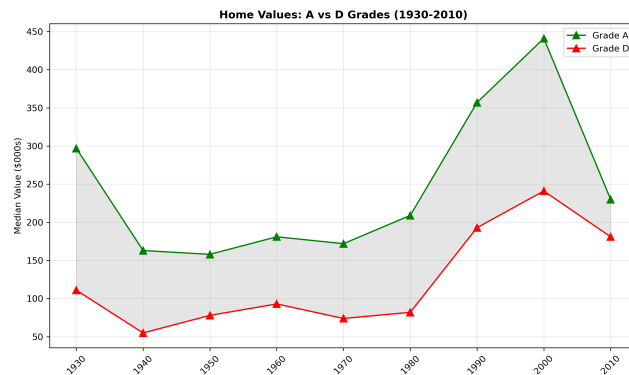
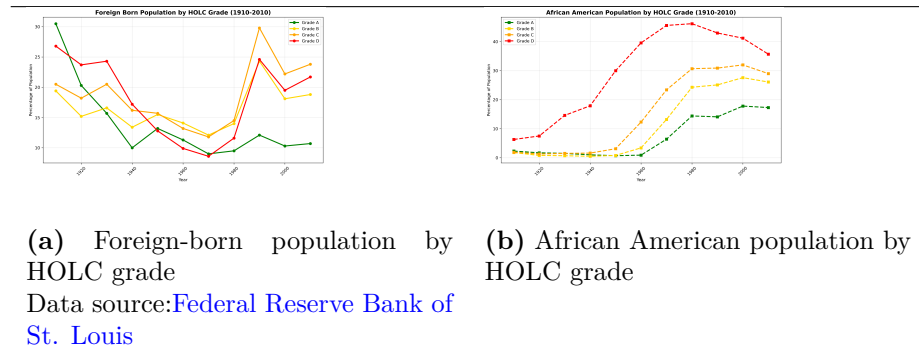


Figure 1: Home value in Grade A and Grade D from (1930-2010) Data
Source: [Federal Reserve Bank of St. Louis](#)

Demographic patterns mirror these housing outcomes. As shown in below, neighborhoods graded C and D are associated with higher proportions of African American and foreign-born residents relative to Grade A areas. These patterns indicate that HOLC grades were not only economic classifications but were also closely aligned with racial and immigrant settlement patterns.



Takeaway: Nationally, redlining is associated with long-lasting differences in both housing values and demographic composition.

3 From National patterns to the Bay Area

While national trends provide important context, they may mask regional variation. We therefore turn to the Bay Area to examine whether similar associations between historical redlining, housing values, and population composition appear at a local scale.

Figure below maps HOLC grades across the Bay Area, highlighting the spatial distribution of historically graded neighborhoods. Areas graded C and D are not evenly distributed; instead, they cluster within specific urban cores, particularly in Oakland, San Jose, and San Francisco, with Red being Grade D and green being Grad A.



Figure 4: HOLC grade by City | *Data Source:* [University of Richmond \[2023\]](#)

To quantify this spatial exposure, The graph below summarizes the proportion of land area classified as Grade C or D within each county. Alameda County exhibits the highest concentration of historically redlined areas, followed by San Francisco and Santa Clara counties. This uneven exposure suggests that counties differ substantially in the extent to which residents may have been affected by redlining policies. I think it is important to note that, even though the proportion of HOLC grades looks similar, the area that it covers in each city is different, with Oakland consist a large area of Grade D, seen in the map above.

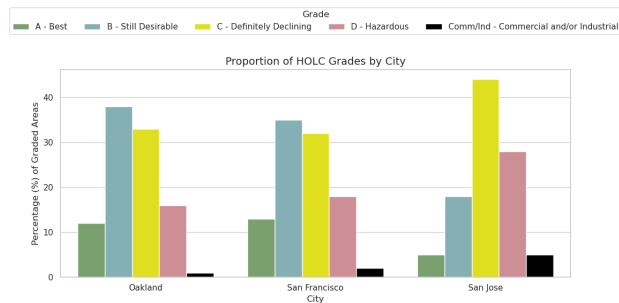


Figure 5: proportion of land area classified as Grade C or D within each county | *Source:* [University of Richmond \[2023\]](#)

4 Contemporary Housing Outcomes in Bayarea (ACS)

We next examine whether counties with greater exposure to historically redlined areas exhibit different contemporary housing outcomes. Figure below shows median home values across Bay Area counties in 2022, with lighter colors indicating higher values and darker colors indicating lower values. Counties with larger proportions of historically graded C and D areas tend to have lower median home values relative to less-exposed counties. San Francisco is in mostly yellow, highest median home value, and East Bay counties in darker colors lower median values.

```
from IPython.display import Image, display
display(Image(filename="figures/ACS/acs_county_homevalue_2022.png", width=700))
```

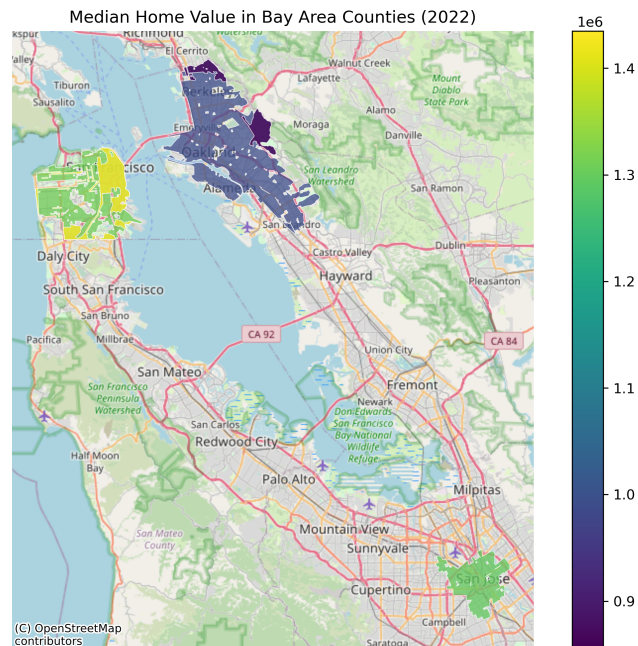


Figure 6: Median Home Value in Selected Bay Area Counties | *Source:* [U.S. Census Bureau](#)

The figure below extends this comparison over time. Although median home values increase across all counties, relative differences persist throughout the period. Counties with greater historical redlining exposure do not appear to catch up to counties with less exposure, suggesting enduring housing value disparities. Alameda County, which includes Oakland, one of our focus of HOLC analysis—illustrates this pattern clearly, exhibiting both a high concentration of historically Grade D areas and comparatively lower median home values than other Bay Area counties.

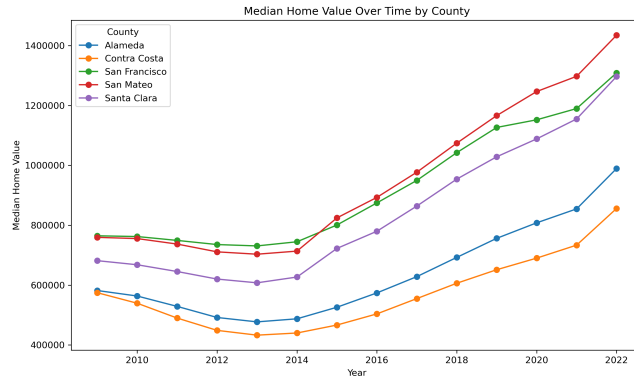
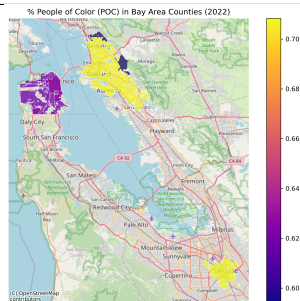


Figure 7: Time-series of median home value by county | *Source:* [U.S. Census Bureau](#)

Key Takeaway: Differences in housing values across Bay Area counties persist over time and align with historical redlining exposure.

5 Demographic Patterns and Change Over Time in Bay Area

Demographic patterns further reinforce these housing disparities. Figure (a) below shows the spatial distribution of People of Color across Bay Area counties, revealing higher proportions in counties with greater historical redlining exposure. With Oakland and San Jose having the high percentage of people of color. Figure (b) below illustrates changes in population composition over time. Counties associated with higher exposure to Grade C and D neighborhoods experience larger increases in the proportion of People of Color, indicating that demographic change has continued to concentrate within historically disfavored areas.



(a) % People of Color (POC) map

(b) Change in % POC over time by county

Data Source: [U.S. Census Bureau](#)

The relationship between housing values and population composition is summarized in Figure 9. Counties with higher proportions of People of Color tend to have lower median home values, consistent with both the national patterns and the historical HOLC classifications.

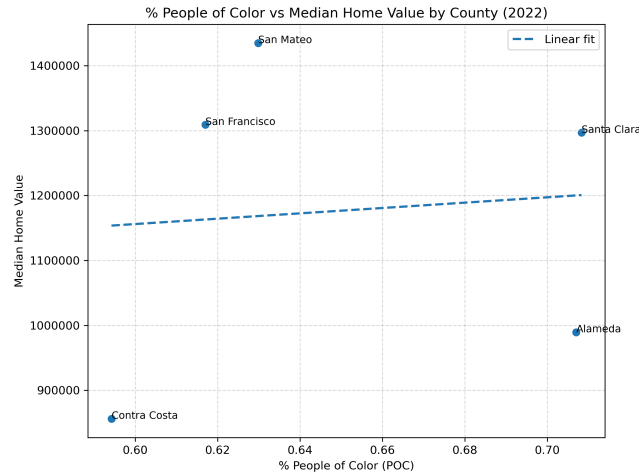


Figure 9: % POC vs median home value in 2022 | *Source:* [U.S. Census Bureau](#)

6 Persistence Over Time — Difference-in-Differences Evidence

To assess whether disparities associated with redlining have narrowed over time, we employ difference-in-differences-style comparisons between historically favored (Grades A/B) and disfavored (Grades C/D) areas. Figure 10 shows parallel trends prior to key policy changes, supporting the use of longitudinal comparison.

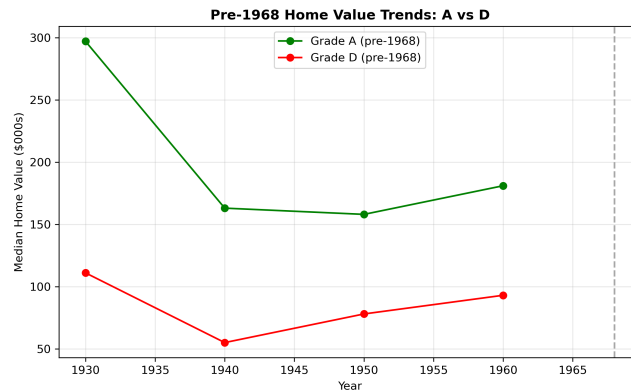


Figure 9: parallel trends prior to key policy change | *Source:* [Federal Reserve Bank of St. Louis](#)

```
import pandas as pd
from src.did_utils import fit_did_ols
```

```
home_long = pd.read_csv("data/home_long.csv")
```

```
did_model_home = fit_did_ols(
    home_long,
    outcome_col="home_value",
    treat_col="treatment",
    post_col="post_1968",
    interaction_col="treat_post",
)
```

```
print(did_model_home.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          home_value    R -squared:                0.517
Model:                  OLS          Adj. R -squared:            0.413
Method:                 Least Squares  F -statistic:              4.990
Date:                  Thu, 18 Dec 2025  Prob (F -statistic):      0.0147
Time:                  00:27:18       Log -Likelihood:          -101.89
No. Observations:      18            AIC:                      211.8
Df Residuals:          14            BIC:                      215.3
Df Model:               3
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	199.7500	39.422	5.067	0.000	115.197	284.303
treatment	-115.5000	55.752	-2.072	0.057	-235.076	4.076
post_1968	82.0500	52.891	1.551	0.143	-31.389	195.489
treat_post	-12.1000	74.799	-0.162	0.874	-172.527	148.327

```
=====
Omnibus:                1.278    Durbin -Watson:                1.585
Prob(Omnibus):           0.528    Jarque -Bera (JB):         1.001
Skew:                   0.535    Prob(JB):                  0.606
Kurtosis:               2.564    Cond. No.                  7.25
=====
```

Notes:

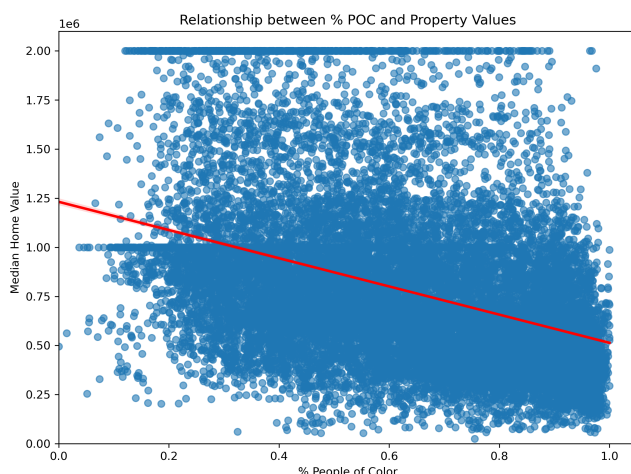
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
/srv/conda/envs/notebook/lib/python3.12/site -packages/scipy/stats/_axis_nan_policy.py:430:
return hypotest_fun_in(*args, **kwargs)
```

The difference-in-differences analysis finds no statistically significant effect

of the 1968 Fair Housing Act on reducing the home value gap between historically redlined (Grade D) and non-redlined (Grade A) neighborhoods. The interaction term (`treat_post`) is small and statistically insignificant, indicating that post-1968 home value trends in redlined areas did not converge toward those of Grade A neighborhoods beyond the pre-existing gap. In practice, this suggests that redlined neighborhoods remained substantially lower in median home value even decades after the policy's enactment. These results imply that although the Fair Housing Act formally prohibited discriminatory lending practices, legislative change alone was insufficient to reverse the long-term economic effects of redlining and the entrenched patterns of disinvestment it produced.

```
display(Image(filename="figures/ACS/acs_regrssion_%poc_value_2.png", width=700))
```



Data source: [U.S. Census Bureau](#)

Regression results indicate that historically redlined areas remain substantially behind non-redlined areas in median home values, with gaps exceeding \$100,000 in recent decades. There is no strong evidence that these gaps have closed over time.

These findings should be interpreted as associational rather than causal; however, they provide consistent evidence that the spatial imprint of redlining persists in contemporary housing outcomes.

7 Conclusion

Together, the national and Bay Area analyses suggest that historical redlining is closely associated with long-lasting housing and demographic disparities. While overall housing values have increased, relative gaps between historically redlined and non-redlined areas remain. These findings highlight how historical policy decisions can shape urban inequality across generations.

8 Limitation and Assumption

This analysis relies on historical HOLC redlining maps and publicly available ACS and housing value data, over which we have no control in terms of original data collection or measurement. We assume that these datasets are internally consistent across time and geography, and that any measurement error is not systematically correlated with treatment status. Given the historical nature of redlining, we treat HOLC grades as fixed indicators of past institutional exposure rather than precise measures of individual-level discrimination.

Our difference-in-differences framework assumes that, absent the policy intervention, treated (historically redlined) and control (non-redlined) areas would have followed parallel trends in housing values. This assumption is partially assessed through pre-policy trend comparisons. While this analysis does not establish causal effects definitively, it allows us to explore whether large-scale anti-discrimination legislation coincided with differential changes in housing outcomes across historically distinct neighborhoods. Given the exploratory scope of the project, we focus on broad, interpretable patterns rather than optimizing model specification or making strong causal claims.

8.1 Author Contribution

Emily Ma:

- Researched data for Historical Redlining (HOLC) and ACS
- Cleaned up both datasets in R
- Created `acs_utils.py` and `did_utils.py`
- Created `test_acs_utils.py`
- Completed `Main.ipynb`
- Completed `README.md`
- Created Binderlink
- Help with `03_acs_data_bay_area.ipynb`
- Created `references.bib`

Larissa Arreola:

- Contributed with the `02_holc_redlining_bay_area.ipynb`
- Built the MyST website and `pdf_builds` directory and its content
- Finalized `ai_documentation` document
- Created `tests_holc_utils.py`

- Translated Emily’s R functions for the geopackage and holc data preprocessing into Python `data_preprocessing.py`
- Created `holc_utils.py`
- Completed Release details of the project requirements (Zenodo, tagging, etc).

Daniel Porter:

- Contributed to the graphs, maps, plots, and accompanying descriptions for the `03_acs_data_bay_area.ipynb` file
- Created a finalized `environments.yml`
- set up license for the GitHub repository
- created binder for the main file

Vittal Vasudevan:

- Created `01_holc_timeseries.ipynb`
- Researched nation wide redlining policy implications and mechanisms for trends
- Created figures and difference-in-difference analysis for nation wide analysis
- Created `.gitignore`
- Created Makefile

References

- Federal Reserve Bank of St. Louis. Redlining-related Economic Data. <https://fred.stlouisfed.org/tags/series?t=redlining>. URL <https://fred.stlouisfed.org/tags/series?t=redlining>. Accessed 2025.
- University of Richmond. Mapping Inequality: Redlining in New Deal America. <https://dsl.richmond.edu/panorama/redlining/>, 2023. URL <https://dsl.richmond.edu/panorama/redlining/>. Accessed 2025.
- U.S. Census Bureau. American Community Survey 5-Year Estimates. <https://www.census.gov/programs-surveys/acs>. URL <https://www.census.gov/programs-surveys/acs>. Accessed 2025.